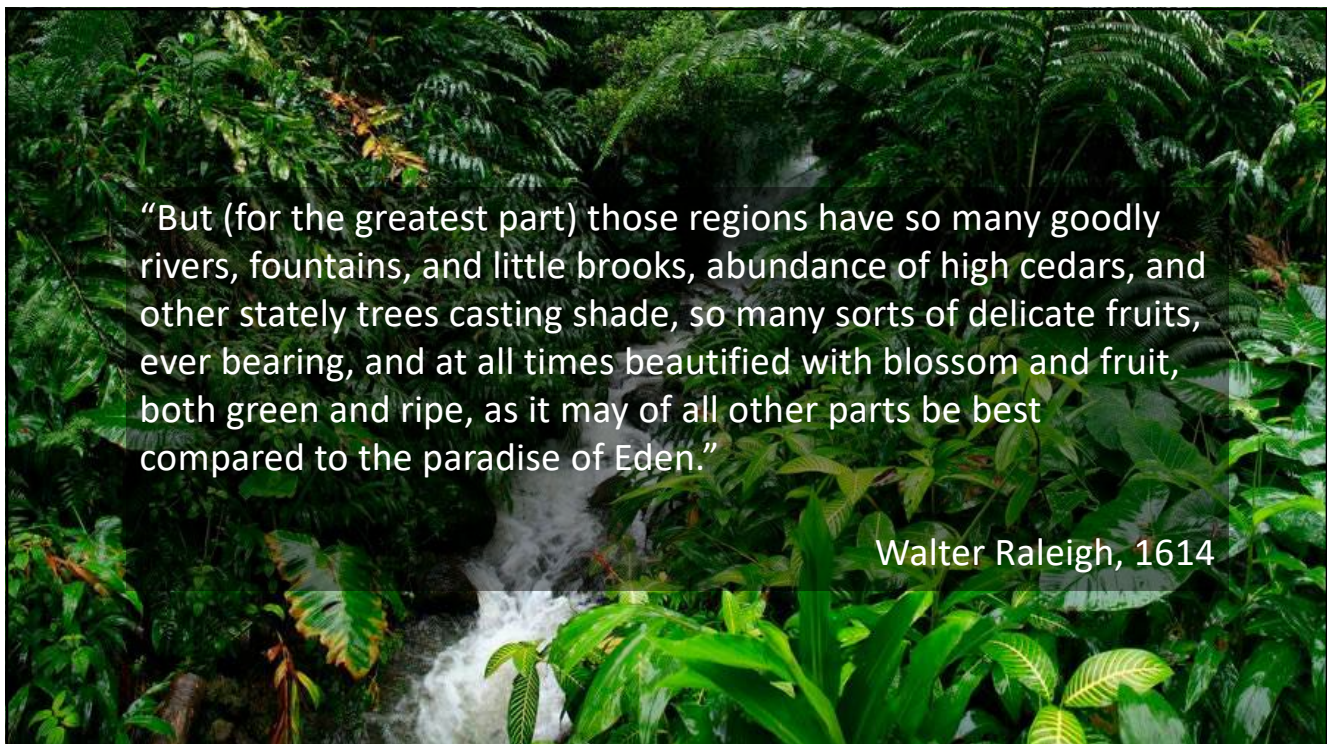




1



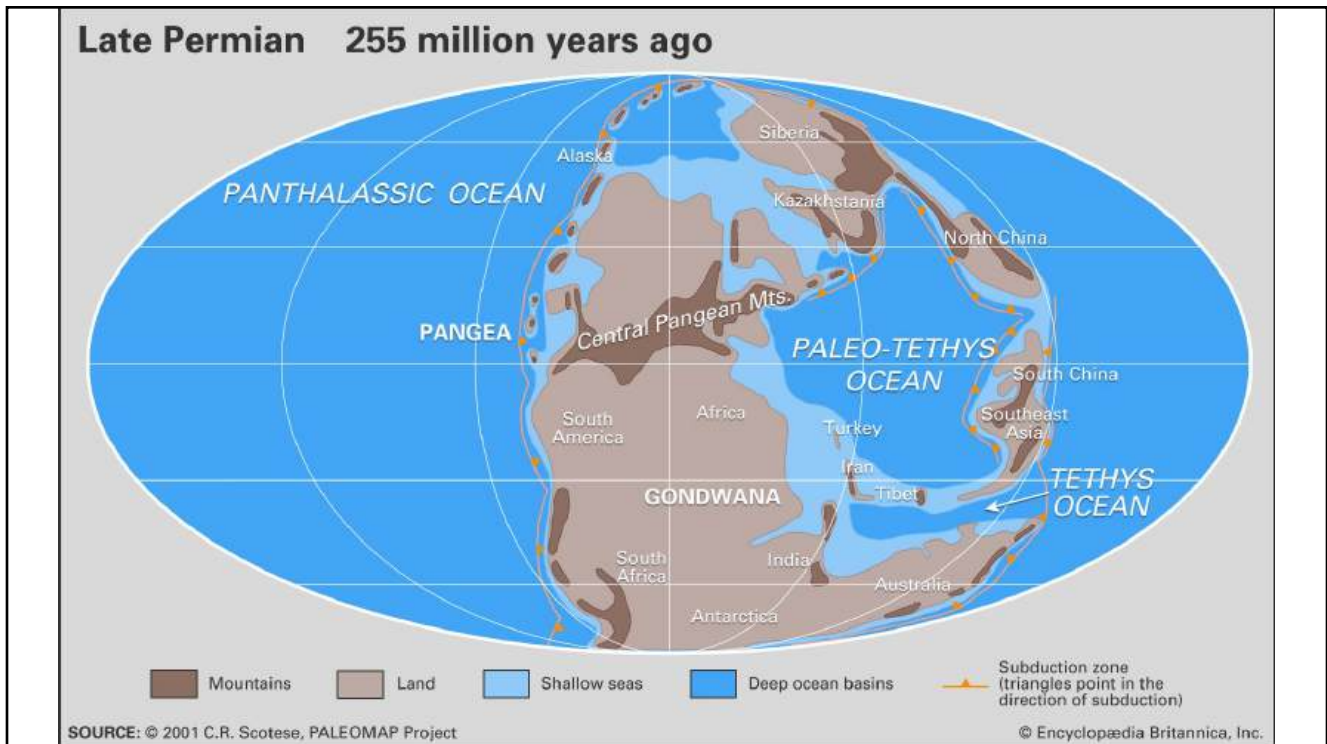
2



3

the
PARADOX
of
LUXURIANCE

4



5



Soil in the Amazon
(and other tropical
regions)...

- 1) Is old
- 2) Is nutrient-poor

6

Soil

Layer of chemically and biologically altered material that overlies rock



Photo: [Soil Science](#) Typic Hapludult (fine-loamy, mixed, semiactive, isohyperthermic) in Loreto, Peru. (CC BY 2.0)

7

organisms at higher trophic levels
get (most of) them from their food



Plants get nutrients from the soil

8

Soil Chemistry

Macronutrients:
Needed in large
amounts.

Micronutrients:
Needed in trace
amounts

TABLE 21.2		
Plant Nutrients and Their Principal Functions		
Nutrients	Principal functions	
Carbon, hydrogen, oxygen	Components of organic molecules	
Nitrogen	Component of amino acids, proteins, chlorophyll, nucleic acids	
Phosphorus	Component of ATP, NADP, nucleic acids, phospholipids	
Potassium	Ionic/osmotic balance, pH regulation, regulation of guard cell turgor	
Calcium	Cell wall strengthening and functioning, ionic balance, membrane permeability	
Magnesium	Component of chlorophyll, enzyme activation	
Sulfur	Component of amino acids, proteins	
Iron	Component of proteins (e.g., heme groups), oxidation–reduction reactions	
Copper	Component of enzymes	
Manganese	Component of enzymes, activation of enzymes	
Zinc	Component of enzymes, activation of enzymes, component of ribosomes, maintenance of membrane integrity	
Nickel	Component of enzymes	
Molybdenum	Component of enzymes	
Boron	Cell wall synthesis, membrane function	
Chlorine	Photosynthesis (water splitting), ionic and electrochemical balance	

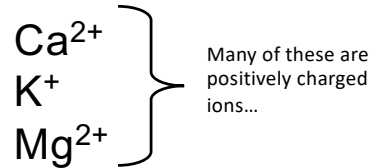
Sources: Salisbury and Ross 1992; Marschner 1995.

ECOLOGY, Table 21.2

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9

Nutrients must be present in simple, water soluble forms
for plant roots to take them up



Nitrogen is the limiting nutrient in the Temperate Zone...

10



11

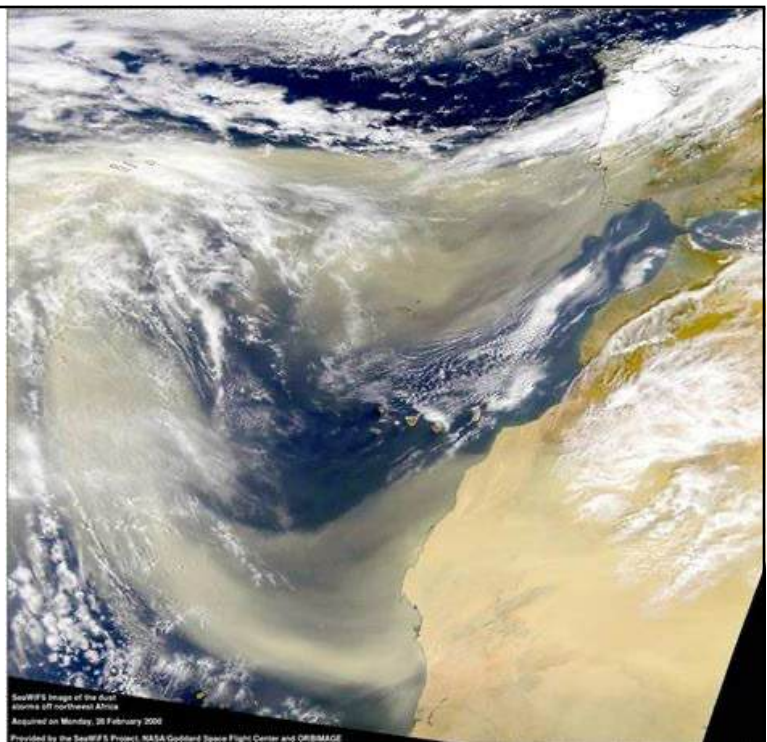


12

When tropical forests are cleared and burned, most nutrients are lost in smoke, ash, and soil erosion.



13

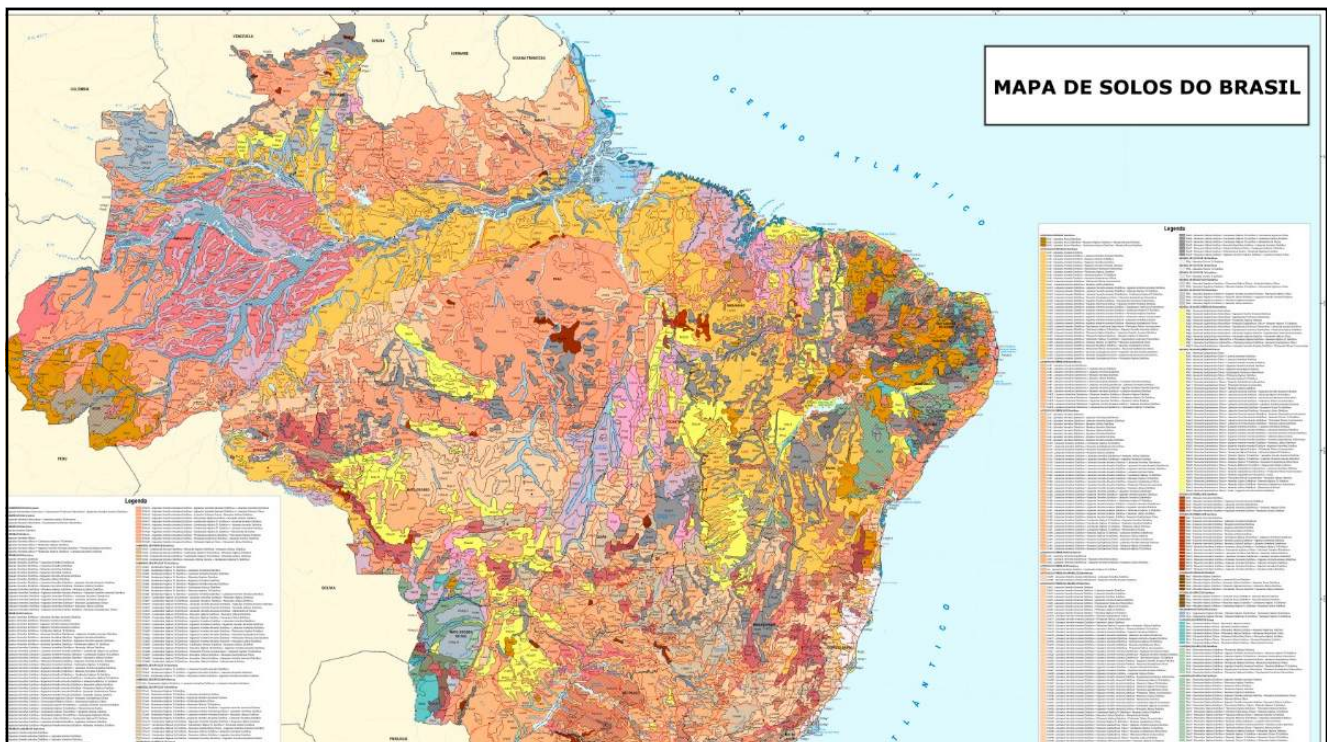


14

Decomposition and rapid nutrient cycling back into living organisms



15



16

Variability in soil quality has major ecological impacts



Plant fertility



Animal abundance,
Home range sizes,

Human settlement and
agriculture



Forest biomass

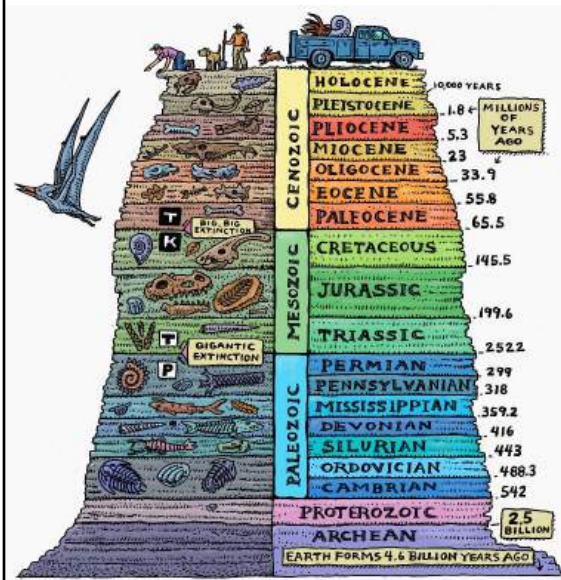


17

Humans have been disturbing
and manipulating tropical forests
for a loooooong time....

18

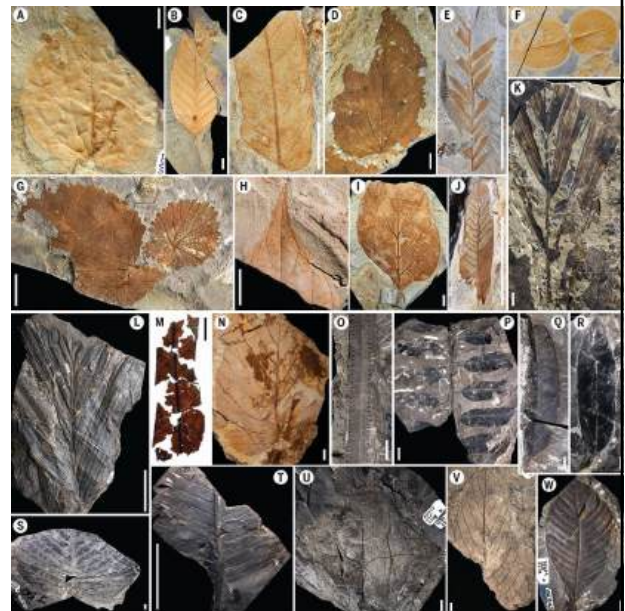
When did humans start living in rain forests?



19

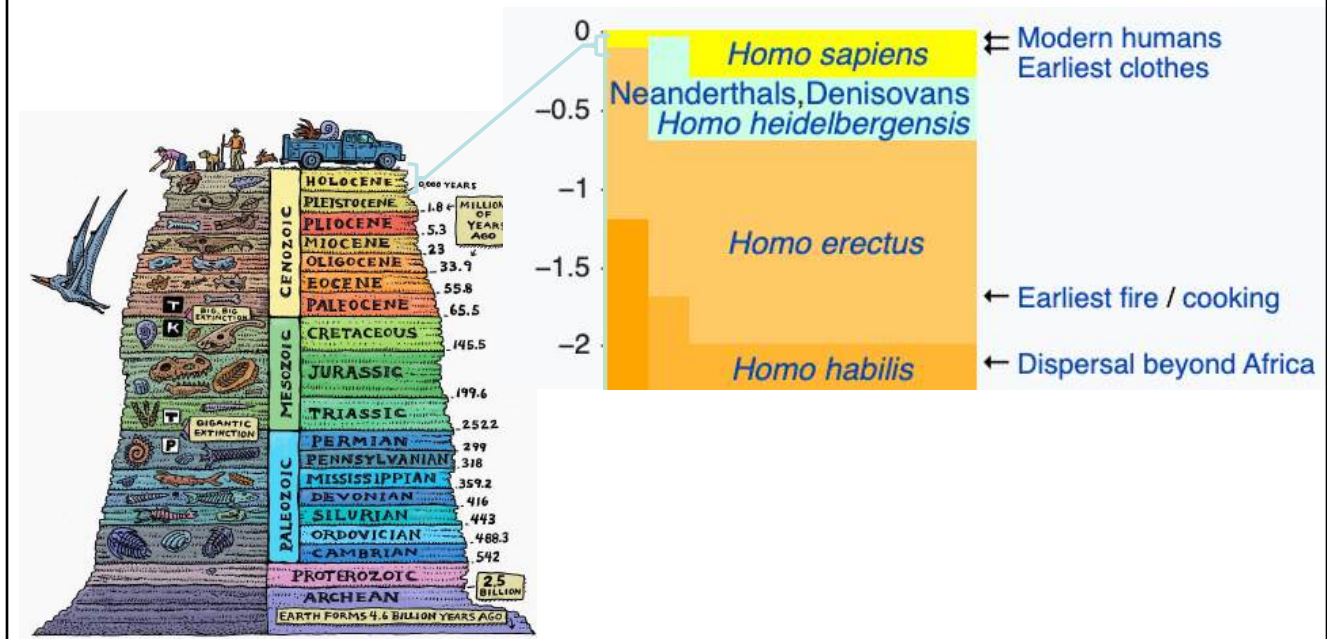
Monica Carvalho, STRI Paleobotanist

6,000 fossil leaves >50,000 grains of fossilized pollen.



20

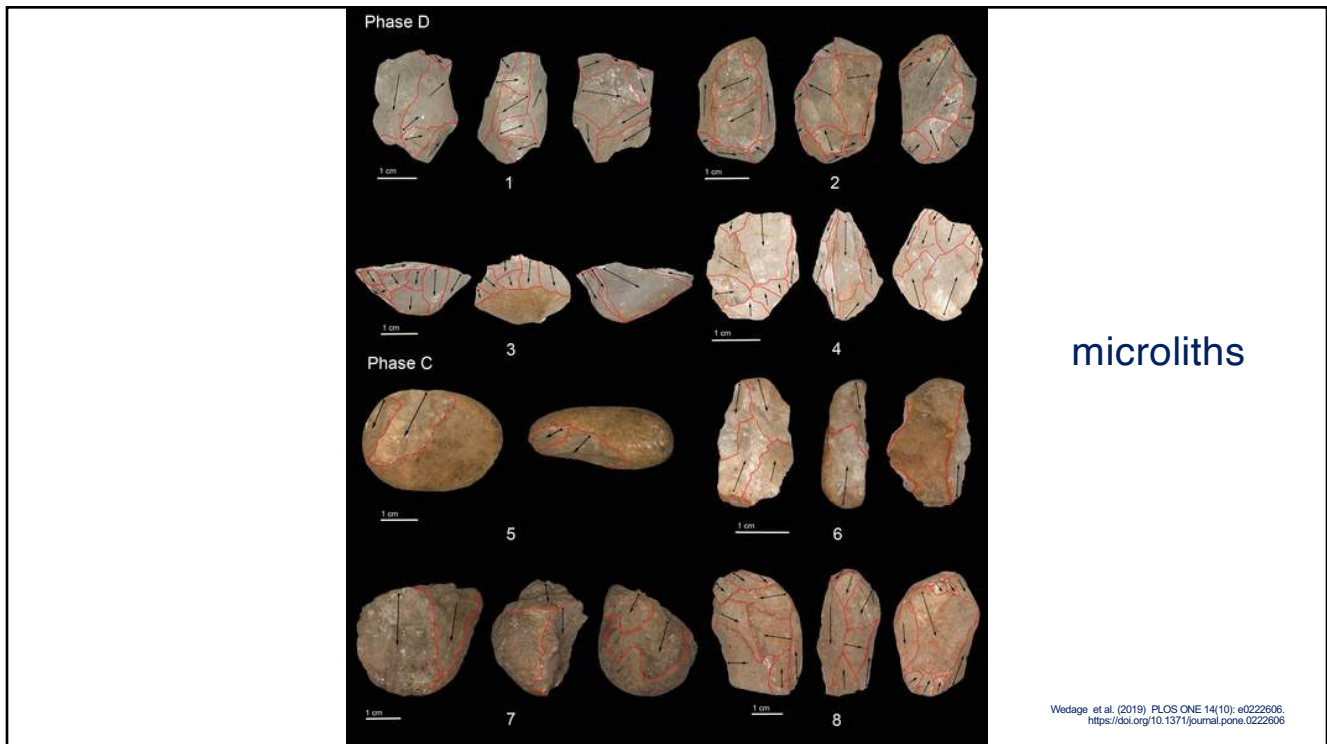
Currently in the Holocene, which began ~11,650 years bp



21



22



23



24

How many people?



25



26



Yavin 4

Star Wars IV: A New Hope

27

1492

Americas: 60 million people

Europe: 70-88 million

28



29



30

Amazonia: 8-20 million



31

how did Amerindians shape rain forests?



Fig. 1 Typical profiles of 'Terra Preta' (a) and Oxisol (b) sites

Amazonian Dark Earths (*Terra Preta do Indio*)

Glaser et al. 2001

32

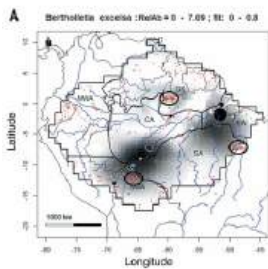
how did Amerindians shape rain forests?

Plant domestication in Amazonia started >8000 B.P.



Dr. Carolina Levis
UFSC-Brazil

Mauritia flexuosa
Euterpe oleracea
E. precatoria
Oenocarpus bataua
Bactris gasipaes



33

Theobroma cacao: 5000 years



maize, manioc, yams

Coffea sp (coffee), *Elaeis guineensis* (oil palm), bananas



34

Ajan Kloss: jungle moon located in the Outer Rim Territories used as a base by The Resistance in The Rise of Skywalker.



35

90%

of the indigenous population
was dead by the early 1600s

(>50 million)

disease
(smallpox, influenza, malaria)
slavery

36

The World's Best Forest Guardians: Indigenous Peoples

Filed Under: [Environment](#)
Last updated October 16, 2018



37

Maya – Chichen Itza

(AD 600-1200)



38

Contemporary Population of Indigenous Peoples in the Americas

~32 million

Peru (45%)

Bolivia (62%)



39

Maya

Ethnolinguistic Group

S. Mexico, Guatemala, Belize, El Salvador, Honduras

~ 7 million people

32 languages



Font size = relative size of speaker population.

Yucatec: 900,000 K'iche': 400,000

100,000–500,000 | 10,000–100,000 | <10,000.

40

200,000

41

LATIN AMERICA

Killings Of Guatemala's Indigenous Activists Raise Specter Of Human Rights Crisis

January 22, 2019 · 6:45 PM ET
Award On All Things Considered

MARIA MARTIN

4-Minute Listen



Mayan indigenous people protest against the government of Guatemalan President Jimmy Morales on the day he gives an address to Congress in Guatemala City, Monday.
Santiago Segura

Fleeing death in Guatemala, two Ixil Maya activists hope to continue their defense of indigenous rights from the US — if they can stay

By [Jasmine Uribe](#) Globe Staff | Updated May 1, 2020, 9:46 a.m.

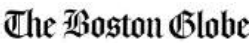


Francisco Chavez Raymundo (left) and Gaspar Cobo Corio were in El Paso, Texas, on April 23. They waited in Mexico for 17 months to get asylum in El Paso. [CHECK OUT THE BEST ON GLOBE](#)

[Leer en español](#) | [Read in Spanish](#)

EL PASO — They left their Ixil Maya town deep in the Cuchumatanes mountains of Guatemala just before dawn and without much time to say goodbye.

Francisco Chavez Raymundo, 45, and Gaspar Cobo Corio, 32, had been part of a tight circle of indigenous activists who in the spring of 2013 helped bring a military dictator to trial over the 1982 genocide of the Ixil people, a Mayan ethnic group that became one of the main targets of systematic racism, rape, and forced displacement during the Guatemalan civil war.

42



Photo: [UN Women](#) (CC BY-NC-ND 2.0)

"55-year-old Candelaria Pec (center, in blue) stands with other women near a communal garden. In addition to growing food for their families, the women use ingredients from their farms and garden to make and market organic shampoo. The women now produce shampoo in bigger batches and in different varieties -- such as aloe, cacao, avocado and honey - - and sell them in local markets.

"I joined this group because I saw the need in my household," says Candelaria Pec. "With the assistance of the project, we have started growing crops, improved our living conditions."